



Development of Data-Driven Cement Demand Forecasting System

By;
Team 3

Multi-Site Case Study

Made with GAMMA

Executive Summary

MIG leadership needs a data-driven demand forecasting system to enable proactive, site-level cement planning. Today's process — rolling 4-week schedules and manual estimates — creates stockouts, overstocking, reactive ordering, and limited supply visibility across sites. This initiative replaces guesswork with predictive intelligence, giving teams the foresight to optimize procurement, reduce waste, and keep projects on schedule.



Business Problem

MIG relies on manual planning and spreadsheet forecasting for cement demand across infrastructure sites. This causes stockouts, overstocking, emergency deliveries, delays, and increased waste.



Business Case

Design forecasting model architectures, establish data pipelines per site, and define documentation standards for reproducible, cross-functional execution.



Impact

The proposed solution is expected to improve operational readiness, reduce stockouts and emergency deliveries, optimize silo capacity and inventory planning, minimize material waste and costs, enhance forecasting visibility, and support sustainability through improved delivery efficiency.



Delivery

Collect forecast accuracy feedback per site, analyze model drift, and recalibrate algorithms for continuous improvement and adaptive demand planning.

Project Charter

Project In scope

- SQLite data extraction
- Data cleaning and validation
- Exploratory data analysis
- Forecasting model development
- Inventory simulation
- Power BI dashboards
- Reorder alert system
- UAT and deployment

Out of scope

- ERP replacement
- Supplier contract renegotiation
- Full enterprise system integration
- Real-time internet of things implementation
- Hardware Installation and upgrades
- Financial budgeting automation
- Realtime GPS logistics tracking

Constraints

- 10-day project timeline
- Dependence on historical data quality
- Multi-site operational variability
- Weather volatility

Assumptions

- Historical consumption data is available
- Planned pours are accurate
- Suppliers respond to forecast-driven orders
- Site inventory reporting is consistent

Sprint Plan

Our multi-site forecasting implementation follows a structured, phased approach that coordinates project managers, data analysts, and data scientists. This methodical strategy minimizes data integration risks, optimizes team resource allocation, and delivers reliable site-level forecasts within a 2-week timeframe.

- 1 — Sprint 1: Data Foundation & Discovery (Day 1-5)**
Project kickoff and stakeholder alignment, all data activities like cleaning, validation, backlog creation, EDA and KPI analysis, forecast feature engineering, sprint review
- 2 — Sprint 2: Forecasting & Dashboard MVP (Day 6-9)**
Develop forecasting models and reporting dashboards
- 3 — Sprint 3: UAT, Deployment & Adoption (Day 9-10)**
Deploy forecasting models across all sites, monitor site-level prediction accuracy, and make iterative model adjustments and UAT execution. Project managers coordinate user training and validate initial forecast integration with procurement workflows..

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Midlands Infrastructure Group (MIG) Ltd. is a Tier-1 UK civil engineering and construction company, founded in Birmingham in 1994.

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Our Forecasting Process Framework

This comprehensive Cement Demand Forecasting Framework enables accurate, site-specific predictions across multiple production and distribution sites. By aligning project managers, data analysts, and data scientists into a unified workflow, we minimize supply risk and maximize operational planning efficiency.



Data Collection & Discovery

Aggregate historical cement consumption data across all sites, identify demand drivers, and align stakeholders on forecasting objectives and success criteria.



Model Design & Planning

Design forecasting model architectures, establish data pipelines per site, and define documentation standards for reproducible, cross-functional execution.



Model Deployment

Deploy forecasting models into production environments, train cross-functional teams, and monitor site-level predictions with regular performance checkpoints.



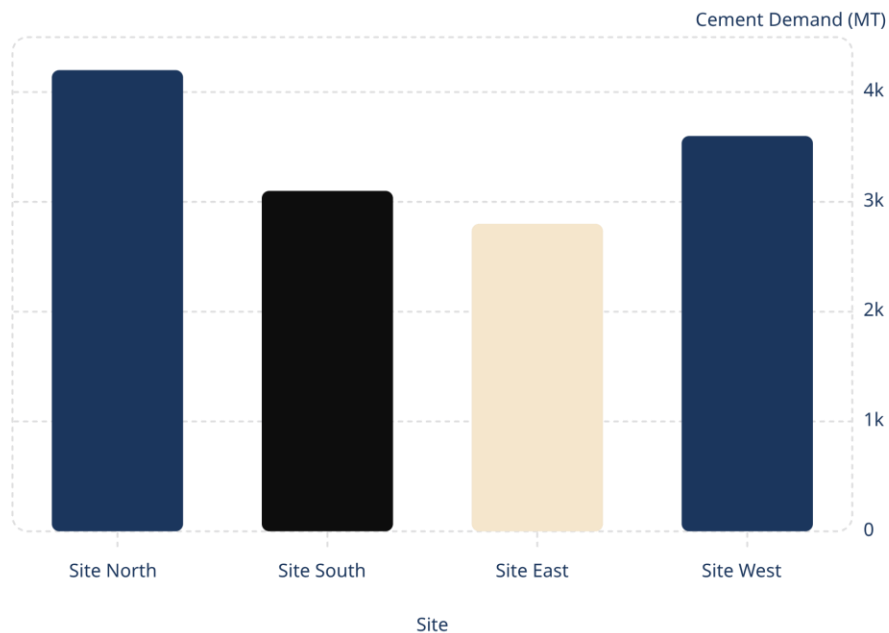
Review & Recalibration

Collect forecast accuracy feedback per site, analyze model drift, and recalibrate algorithms for continuous improvement and adaptive demand planning.

Forecasting Performance Metrics

These key metrics highlight the accuracy and operational impact of our multi-site cement demand forecasting models. Rigorous tracking enables data-driven decisions by project managers, analysts, and data scientists for continuous model improvement.

- **Forecast Accuracy (MAPE):** Mean Absolute Percentage Error across sites measures how closely predictions match actual cement consumption volumes.
- **Inventory Optimization Score:** Reflects reduction in overstock and stockout events achieved through more precise demand signals per site.
- **Model Cycle Time:** Measures efficiency gains in retraining and redeployment pipelines, indicating decreased time from data refresh to updated forecast.
- **Stakeholder Adoption Rate:** Gauges the degree to which project managers and site operators actively use forecasts in procurement and scheduling decisions.
- **Cross-Site Consistency:** Evaluates model performance parity across geographically distributed sites to ensure equitable planning support.



Executive Dashboard Insights

- Provides a high-level view of performance across all sites and regions.
- Displays total planned pours, deliveries, and consumption, helping leadership monitor efficiency.
- Shows balanced cement type usage (CEM I, II, III), indicating standardized procurement and production.
- Monthly consumption trends reveal seasonal fluctuations; dips in February and September suggest downtime or supply delays.
- Highlights top sites with pour disruption and overcapacity, guiding executives toward operational improvements.
- Enables leadership to identify high-risk sites, allocate resources effectively, and align operations with the forecasting model.



Consumption by Cement Type



30

Total Sites

1.01M

Total Planned Pours

963.15K

Total Deliveries Vo...

779.93K

Total Consumed T...

9.1%

Idle %

24.1%

Pour Disruption %

Consumed Tones by Month



site_id	Stockout %	Overcapacity %	Waste Risk %	Pour Disru
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site_id	Stockout %	Overcapacity %	Waste Risk %	Pour Disru
SITE_022	51.4%	0.0%	4.20%	
SITE_008	49.9%	0.0%	4.38%	
SITE_011	49.8%	0.0%	4.11%	
SITE_021	48.8%	0.0%	3.92%	
SITE_020	48.4%	0.0%	5.84%	
SITE_017	47.9%	0.0%	5.20%	
SITE_010	47.3%	0.0%	4.84%	
SITE_005	47.1%	0.0%	5.57%	
SITE_025	47.0%	0.0%	5.29%	
SITE_001	46.5%	0.0%	5.57%	
SITE_007	46.3%	0.0%	4.84%	
SITE_030	45.9%	0.0%	5.38%	
SITE_018	45.4%	0.0%	4.74%	
SITE_003	45.3%	0.0%	5.66%	
SITE_006	13.4%	0.0%	20.89%	
SITE_016	10.0%	15.3%	19.98%	

Top 10 Sites by Pour Disruption

SITE_022	51.4%
SITE_008	49.9%
SITE_011	49.8%
SITE_021	48.8%
SITE_020	48.4%
SITE_017	47.9%
SITE_010	47.3%
SITE_005	47.1%
SITE_025	47.0%
SITE_001	46.5%

Top 10 Overcapacity Sites

SITE_015	99.5%
SITE_012	99.1%
SITE_023	99.0%
SITE_002	98.8%
SITE_009	98.8%
SITE_027	98.7%
SITE_029	98.3%
SITE_004	98.1%
SITE_019	98.1%
SITE_013	50.8%

Site Manager Dashboard Insights

- Offers a localized view of each site's performance and customer behaviour.
- Compares planned vs actual pours, deliveries vs consumption, and silo capacity utilization.
- Regional behaviour analysis:
 - East → highest consumption and chaotic patterns.
 - South → most aggressive and volatile.
 - West → balanced and moderate.
 - North → stable and predictable.
- Row-Level Security (RLS) ensures each manager sees only their site data.
- Helps site managers anticipate demand, reduce waste, and align operations with the forecasting model.



1.01M

Total Planned Pours

963.15K

Total Deliveries Volume

779.93K

Total Consumed Tones

7,950

Stockout Days

11,439

Overcapacity Days

2,983

Idle Silo Days

81.67

Consumption Variability

Consumption by Cement Type



Planned vs Actual Pours



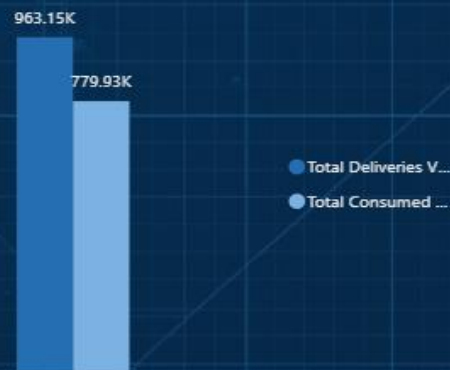
Regional Consumption by Customer Behaviour



Planned Pours vs Silo Capacity



Deliveries vs Consumption (Delivery Sufficiency)



RECOMMENDATIONS

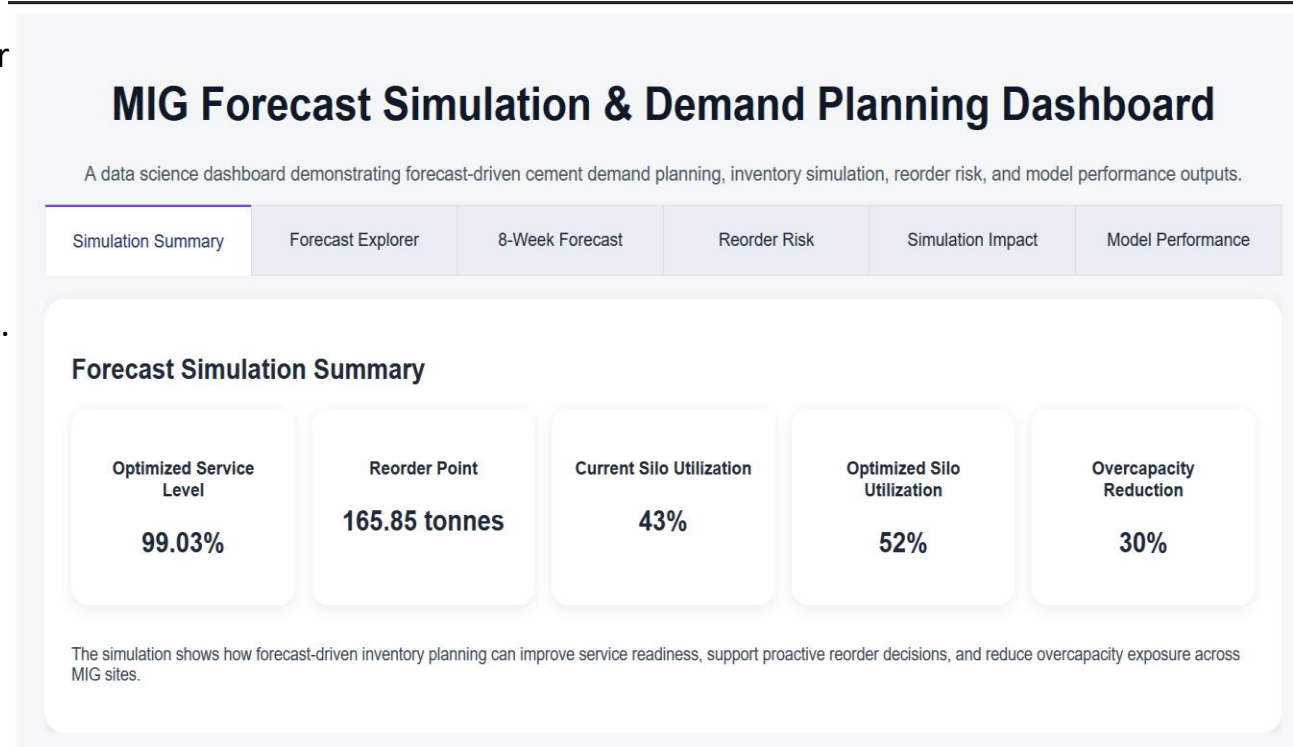
To maximize the value of these dashboards and ensure seamless integration with the forecasting model:

- Strengthen forecasting for chaotic and aggressive regions (East and South).
- Align deliveries with actual consumption to reduce waste and overcapacity.
- Use silo utilization data to optimize storage and delivery timing.
- Encourage regular dashboard use for performance tracking and decision support.
- Train site managers to interpret behavioural patterns and adjust operations accordingly.

Together, these dashboards prepare MIG's data environment for forecasting integration, enabling improved demand planning, reduced waste, and enhanced delivery efficiency across all sites.

FORECAST SIMULATION DASHBOARD

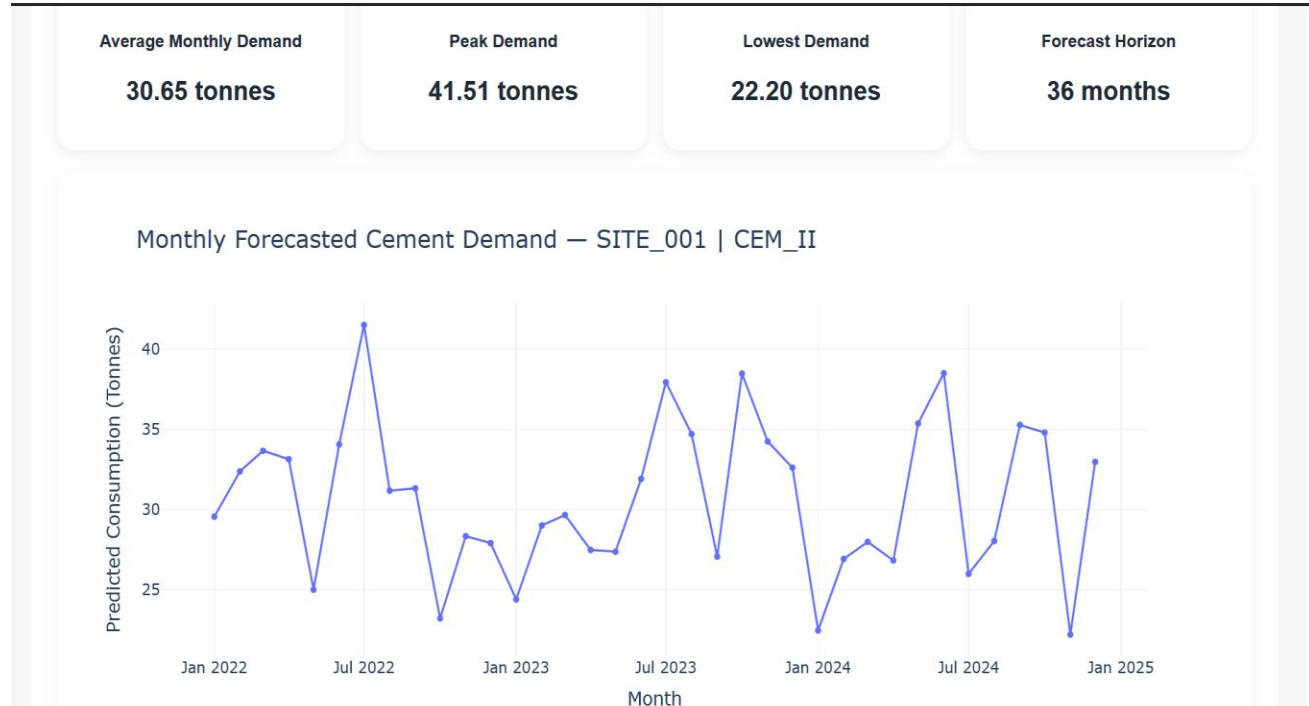
- Developed an interactive forecasting analytics dashboard for multi-site cement demand planning.
- Integrated forecasting outputs, inventory simulation, and reorder monitoring into a unified interface.
- Supports operational visibility across sites and cement categories.
- Successfully deployed using Render cloud hosting.



Render deployment link:- <https://mig-forecast-simulation-dashboard.onrender.com/>
GitHub repo:- <https://github.com/Laurel-hub/mig-forecast-simulation-dashboard>

FORECAST EXPLORER & DEMAND FORECASTING

- Dynamic site and cement type forecast filtering.
- Monthly demand forecasting across a 36-month horizon.
- KPI monitoring for average, peak, and minimum demand.
- Supports proactive procurement and inventory planning.



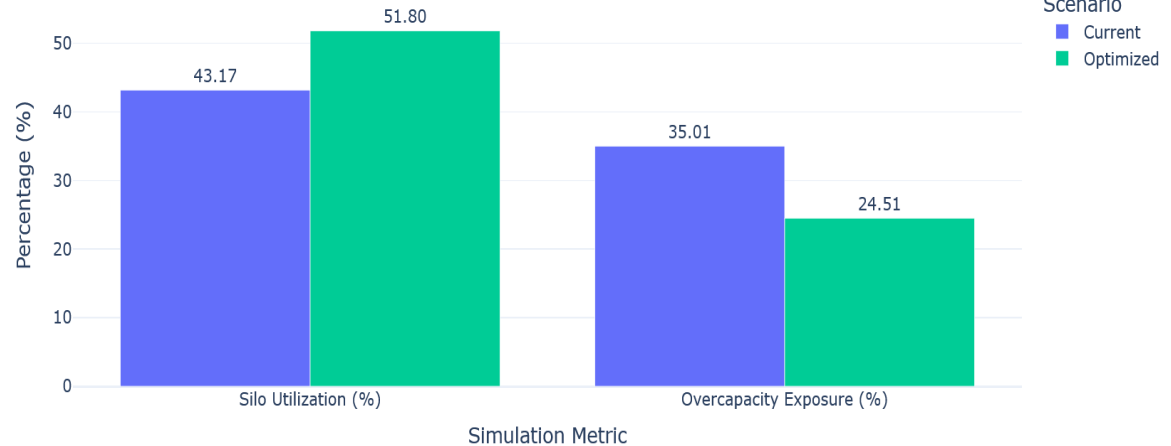
INVENTORY SIMULATION & OPTIMISATION IMPACT

- Simulation compares current and optimized inventory performance.
- Forecast-driven optimization improves silo utilisation efficiency.
- Reduced overcapacity exposure supports operational cost control.
- Reorder alert outputs support proactive replenishment planning.

Inventory Efficiency and Simulation Impact

This section visualises the operational impact of inventory optimization by comparing current and optimized inventory performance.

Current vs Optimized Inventory Simulation Impact



FORECAST MODEL EVALUATION & DEPLOYMENT

- Compared multiple forecasting approaches for predictive accuracy.
- SARIMAX selected as preferred forecasting model based on lower MAPE.
- Dashboard deployed as a cloud-accessible analytics solution.
- Enables scalable forecasting visualization and stakeholder accessibility.

Forecasting Model Performance

Model comparison results validate forecasting accuracy and support the selection of SARIMAX as the preferred forecasting approach.

Forecast Model Performance Comparison

